

Ravi Sharma

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SUMMARY

Full-Stack & Applied AI Engineer specializing in scalable backend architecture and production-grade RAG systems. Proven experience in optimizing system latency, deploying vector databases, and engineering reliable, high-performance software.

EXPERIENCE

Full-Stack Engineer

March 2025 – May 2026

Strengthverse Private Limited

- Delivered 3 production-ready full-stack features (User Settings, Feedback, Sports Articles) on a React Native + Node.js/Express monolithic MVC application, engineered to support 350+ concurrent users on these features.
- Optimized backend services using Redis and BullMQ powered backend services with MongoDB and REST APIs, reducing latency by 40–60% while delivering secure asynchronous processing with sub-300 ms response times.
- Contributed to scaling the overall platform's supported user capacity from 50,000 to 800,000+ and concurrent capacity from 100 to 1,000+ users through high-quality feature engineering and production-focused development.

PROJECTS

Nexdoc - Offline AI Doc Search | *Python, FastAPI, LanceDB, Electron* [GitHub](#) | [Demo Video](#) | [Nexdoc.exe](#)

- Architected a producer-consumer ingestion pipeline utilizing Watchdog and an auto-scaling, CPU-aware worker pool, enabling non-blocking concurrent processing across 5 document formats.
- Engineered modular extraction workflows integrating text normalization and sentence-aware chunking, achieving 88% extraction accuracy while persisting embeddings and enriched metadata into LanceDB and SQLite.
- Accelerated query latency from 5 seconds to under 1 second via LanceDB vector search, while maximizing retrieval accuracy through a multi-stage hybrid filtering, re-ranking, and dynamic context-windowing pipeline.
- Deployed a local-first desktop application featuring an optional Gemini-powered AI mode with strict citation grounding, anchoring 100% of synthesized answers directly to the top-5 verifiable document chunks.

FishPay - Payment Management System (Baas) | *Spring Boot, PostgreSQL* [GitHub](#) | [Demo Video](#) | [Demo Apk](#)

- Built a plug-and-play payment infrastructure with 8 core payment management features for businesses by implementing Razorpay webhook-based server-to-server reconciliation with signature verification for secure and consistent payment processing.
- Reduced payment confirmation latency from ~3 seconds to under 750 ms by returning the payment response immediately while asynchronously generating, uploading, and persisting invoice PDFs in the background.
- Containerized the backend using a multi-stage Docker build, 1st compiling the application with Maven and JDK, then packaging only the generated .jar into the runtime image to reduce image size and improve security.

Stamin - AI Personal Health & Fitness Coaching Platform | *Node.js, Python* [GitHub](#) [Architecture](#)

- Constructed dual-backend architecture using Node.js and FastAPI for scalable AI orchestration and inference
- Designed multimodal ingestion pipelines for PDF, DOCX, and image processing using OCR, CLIP, and BLIP
- Implemented preprocessing mechanisms with validation, metadata extraction, PII masking, and semantic chunking
- Developed asynchronous RAG system using BullMQ, Redis, vector embeddings, Pinecone, and MongoDB
- Designed layered query validation with moderation, intent classification, risk scoring, and routing workflows
- Programmed REST APIs for health data ingestion and personalized AI-driven guidance logic

TECHNICAL SKILLS

Languages: Java, JavaScript, Python, SQL

Frontend: React Native, React.js, Next.js, Tailwind CSS, Redux Toolkit, Axios, Zod, TypeScript, HTML5, CSS

Backend: Spring Boot, Node.js, Express.js, FastAPI, REST APIs, Spring Security, Hibernate, JPA, JWT, bcrypt

Databases: MongoDB, MySQL, PostgreSQL, Pinecone, FAISS, LanceDB

Tools & Platforms: Git/GitHub, Docker, Postman, Maven, Render, Electron, Redis, BullMQ, AWS S3, AWS EC2

Concepts: Data Structures & Algorithms, System Design, RAG, Multimodal AI, Event-Driven Architecture

EDUCATION

Maulana Abul Kalam Azad University of Technology (MAKAUT)

Kolkata, India

Bachelor of Computer Applications (BCA)

August 2021 – May 2024

Techno India University

Kolkata, India

Master of Computer Applications (MCA)

August 2024 – June 2026